

^dAlthough the barbell biceps curl exercise is a “pulling” movement and occurs before another “pull” exercise (shoulder shrug), it does not affect the athlete’s ability to perform the shoulder shrug exercise.

technique in structural or standing exercises.

Step 5: Training Load and Repetitions

Load most simply refers to the amount of weight assigned to an exercise set and is often characterized as the most critical aspect of a resistance training program (50, 106, 162, 163).

Terminology Used to Quantify and Qualify Mechanical Work

Mechanical work can be defined as the product of *force* and *displacement* (sometimes referred to as *distance*). An athlete can perform (external) mechanical work via demands made on the body to generate (internal) metabolic energy. Thus, it is important to quantify the amount of mechanical work or degree of metabolic demand to plan variation in the training program and to avoid the exhaustion phase of Selye's General Adaptation Syndrome associated with overtraining (33, 148).

A quantity measure for resistance training work is needed. Traditionally, at least in the sport of weightlifting, this work is called the *load*, and one can calculate it by multiplying each weight lifted by the number of times it is lifted and summing all such values over a training session. **Volume-load** is a proxy for work, because displacement and force production are not measured in applied settings. However, *volume-load* (90) is a better term than just *load*. This quantity is related to mechanical work and the associated metabolic energy demands and physiological stress and is distinguished from **repetition-volume** (i.e., the total number of repetitions; see Step 6: Volume for more explanation).

Volume-load should be considered as *system mass volume-load* in the calculation of resistance training in which the athlete or a mass is moved (e.g., loaded jump squats). For example, an 80-kilogram athlete with a 40-kilogram jump squat load for 4 sets of 3 is doing 120 kilograms $\times$ 12, or 1,440 kilograms. In contrast, when the athlete's mass is not being lifted (e.g., bench press or shoulder press), only the external load should be included in the volume-load calculation. Volume-load approaches are also useful in quantifying the nature of the total resistance training load, by separating the volume-load from core and assistance exercises or delineating between hypertrophy, maximal strength, and power training. In this way, the strength and conditioning professional can plan or determine not just the total volume-load for the session but also what stimulus is achieved primarily from the session.

The typical calculation of volume-load (i.e., sets $\times$ repetitions $\times$ load) does not account for the distance that external load is traveling. Given that athletes possess unique anthropometric characteristics (e.g., height and arm length), the

amount of work being performed by athletes within a training session may vary. For example, athletes who are 5 feet 6 inches (1.68 m) and 6 feet (1.83 m) in stature who perform the back squat exercise with the same load and to the same squat depth are performing different magnitudes of work based on the displacement of the barbell. Within this scenario, the taller athlete must move (displace) the external load a greater distance, which results in performing more work than the shorter athlete. When monitoring training load, load displacement can be accounted for by calculating **volume–load displacement**, which multiplies volume–load by the displacement of the exercise. Although strongly correlated to volume–load, researchers have indicated that volume–load displacement may provide a better indicator of an athlete’s overall workload totals (78). Thus, if strength and conditioning professionals have the time to measure the displacements of exercises for their athletes (using appropriate technology), greater insight into how athletes respond to various training phases may be provided.

Note that the volume–load is not affected by the set and repetition scheme (e.g., 15 sets of 1 repetition, 5 sets of 3 repetitions, 3 sets of 5 repetitions, or 1 set of 15 repetitions). Various set and repetition schemes affect the true **intensity** value for resistance exercise and indicate the *quality* of work performed. Instead of using time to calculate mechanical or metabolic power or intensity, it is more practical to use a value that is proportional to time—namely, rep-volume. The more repetitions performed, the longer the training session; rest period lengths are an additional consideration and are not directly accounted for. Dividing volume–load by rep-volume results in the average weight lifted per repetition per workout session (162).

Relationship Between Load and Repetitions

The number of times an exercise can be performed (**repetitions**) is inversely related to the load lifted; the heavier the load, the lower the number of repetitions that can be performed. Therefore, focusing on one training goal automatically implies the use of a certain load and repetition regimen (e.g., training for maximal muscular strength involves lifting very heavy loads [i.e., $\geq 80\%$ 1RM] for few repetitions [i.e., ≤ 6 repetitions]).

Before assigning training loads, the strength and conditioning professional should understand this relationship between loads and repetitions. Load is commonly described as either a certain percentage of a **1-repetition maximum (1RM)**—the greatest amount of weight that can be lifted with proper technique for only 1 repetition—or the most weight lifted for a specified number of repetitions, an **xRM**. For instance, if athletes can perform 10 repetitions with 60 kilograms in the back squat exercise, the 10RM is 60

kilograms. It is assumed that the athlete provided a maximal effort; if he or she had stopped at 9 repetitions but could have performed 1 more, a 10RM would not have been achieved. Likewise, if he or she lifted 55 kilograms for 10 repetitions but could have performed more, the true 10RM was not accurately assessed because the athlete possibly could have lifted 60 kilograms for 10 repetitions.

[Table 18.8](#) shows the relationship between a submaximal load—calculated as a percentage of the 1RM—and the number of repetitions that can be performed at that load. By definition, 100% of the 1RM allows the athlete to perform 1 repetition. As the percentage of the 1RM (i.e., the load lifted) decreases, the athlete will be able to successfully complete more repetitions. However, strength and conditioning professionals should be cautious because the number of repetitions that can be performed at a given load may be specific to the exercise as well as the individual (120, 134).

TABLE 18.8 Percent of the 1RM and Repetitions Allowed (%1RM–Repetition Relationship)

%1RM	Number of reps allowed
100	1
95	2
93	3
90	4
87	5
85	6
83	7
80	8
77	9
75	10
70	11
67	12
65	15

Note: The number of reps performed at each percentage may be based on individual characteristics as well as the exercise being performed (120, 134).

Data from Hoeger et al. (1987); Hoeger et al. (1990); Mayhew et al. (1992).

Although %1RM–repetition tables provide helpful guidelines for assigning an athlete’s training loads, research to date does not support the widespread use of such tables for establishing training loads for every exercise assigned to athletes, for the following reasons:

- [Table 18.8](#) assumes there is a linear association between the loads lifted and the repetitions performed; however, several studies have reported a curvilinear relationship (95, 99, 103).
- Resistance-trained athletes may be able to exceed the number of repetitions listed in the table at any given percentage of their 1RM, especially in lower body core exercises (74, 75, 120, 134).
- The number of repetitions that can be performed at a certain percentage of the 1RM is based on a single set. When an athlete performs multiple sets, the loads may need to be reduced so that the desired number of repetitions can be completed in all the sets. Despite the prevalence of 1RM research, athletes may not always perform the predicted number of repetitions at a specified percentage of a 1RM. For instance, researchers have shown that subjects were able to perform two or three *times* more repetitions than are listed in [table 18.8](#) (74, 75, 134).
- A certain percentage of the 1RM assigned to a machine exercise can result in more repetitions at the same percentage of the 1RM than with a similar free weight exercise (74, 75).
- Exercises involving smaller muscle areas may not produce as many repetitions as seen in [table 18.8](#), and exercises recruiting large muscle areas are likely to result in more repetitions performed (183).
- Empirical evidence further suggests that as the percentage of the 1RM decreases, the variability in the number of repetitions that can be completed increases. For example, the most accurate relationship between percentages of the 1RM and the maximum repetitions possible is for loads greater than 75% of the 1RM and fewer than 10 repetitions (133) with greater accuracy coming with even lower repetitions (45).

Therefore, loads calculated from the %1RM in [table 18.8](#) should be used only as a guideline for estimating a particular RM load for a resistance training exercise. Even with the inherent weaknesses just explained, it appears that it is still more accurate to assign loads based on a percentage of a test-established 1RM than it is to estimate a 1RM from a submaximal load (74, 75).

1RM and xRM Testing Options

To gather information needed to assign a training load, the strength and conditioning professional has the option of determining the athlete’s

- actual 1RM (directly tested),
- estimated 1RM from an xRM test (e.g., a 10RM), or
- xRM based on the number of repetitions planned for that exercise (the “goal” repetitions; e.g., 5 repetitions per set).

Once the actual 1RM is measured or estimated, the athlete’s training load is calculated as a percentage of the 1RM. Alternatively, a xRM test may be performed based on goal repetitions, thereby eliminating computations or estimations. In many cases, the strength and conditioning professional will use a variety of testing options depending on the exercises selected and the athlete’s training background. A common strategy for testing sufficiently conditioned athletes is to conduct a 1RM test in several core exercises and use xRM testing for assistance exercises.

Testing the 1RM

To assign training loads based on a percentage of the 1RM, the strength and conditioning professional must first determine the athlete’s 1RM. This method of assessment is typically reserved for resistance-trained athletes who are classified as intermediate or advanced and have exercise technique experience in the exercises being tested; however, 1RM testing can be safely performed with youth (19), adolescents (49), inexperienced adults (24), and older adults (127). Individuals who are untrained, inexperienced, injured, or medically supervised may not be appropriate participants for 1RM testing. One-repetition maximum testing requires an adequate training status and lifting experience, because the assessment of maximal strength places significant stress on the involved muscles, connective tissues, and joints, although this has been shown to be appropriate and reliable in youth and collegiate athletes with approximately 6 months’ experience in the power clean and back squat (24, 49). However, it may be more appropriate to complete a 3RM test instead of a maximal 1RM test because ignoring an athlete’s training status and exercise technique experience diminishes the safety and accuracy of 1RM test results.

When selecting exercises for 1RM testing, the strength and conditioning professional should choose core exercises, because the large muscle groups and multiple joints are better able to handle the heavy loads. Despite this guideline, an exercise should not be selected for 1RM testing if it cannot provide valid and reliable data (i.e., does not accurately and consistently assess maximal muscular strength). For instance, the large upper back musculature and multiple joints involved in the bent-over row exercise can probably tolerate the loads for a 1RM test, but maintaining a correct body position throughout testing would be extremely difficult. The weaker stabilizing muscles of the lower back might

become very fatigued after several testing sets, resulting in a loss of proper exercise technique and an invalid test.

A variety of procedures can be used to accurately determine a 1RM; one method is described in [figure 18.8](#). Despite an orderly testing sequence, variations in training status and exercise type will affect the absolute load increases in sequential testing sets. For example, the gradual load increase for 1RM attempts for an athlete who can back squat 495 pounds (225 kg) may be 20 to 30 pounds (9-14 kg) per testing set. For a weaker athlete with a back squat 1RM of 100 pounds (45 kg), a 20- or 30-pound testing load increment is too aggressive and is not precise enough to yield an accurate 1RM value. To improve the appropriateness and accuracy of the sequential testing sets, [figure 18.8](#) also includes relative percentages that can be used instead of the absolute load adjustments.

Estimating a 1RM

When maximal strength testing is not warranted, testing with a 3RM or 5RM load (and then estimating or predicting the 1RM) can be a suitable secondary option. This approach is appropriate for nearly all athletes, provided they can demonstrate the proper technique in the exercise tested. Core and assistance exercises can be selected for RM testing, but excessive warm-up and testing sets may fatigue the athlete and compromise the accuracy of the test. It should be noted, however, that the level of accuracy of predicting a 1RM from other xRM tests becomes less accurate (i.e., commonly resulting in overestimations of the 1RM) as the number of repetitions completed increases and may also be affected by the type of training with which the athlete is familiar (134). Additionally, power exercises do not lend themselves well to xRM testing above 5 repetitions for repeated testing sets because technique can deteriorate rapidly (71). Lower (and more accurate) xRM determinations using heavier loads can be made once the athlete has sufficient training and technique experience.

The protocol for multiple RM testing is similar to that for 1RM testing, but each set requires multiple repetitions, not one. After the completion of warm-up sets, the athlete's sequential load changes for the 10RM test are smaller than those listed in [figure 18.8](#) (approximately one-half). Continue the process of testing until a load allowing the selected number of repetitions is determined. An experienced strength and conditioning professional will be able to adjust the loads so that the multiple RM can be measured within 3 to 5 testing sets.

Using a 1RM Table To estimate the athlete's 1RM, consult [table 18.9](#). In the "Maximum repetitions (RM)" = 10 (%1RM = 75) row, first find the tested 10RM load; then read across the row to the "Maximum repetitions (RM)" = 1 (%1RM = 100) row to discover the athlete's projected 1RM. For example, if an athlete's

10RM is 300 pounds, the estimated 1RM is 400 pounds. As noted in connection with [table 18.8](#) ([page 544](#)), the %1RM–repetition associations vary in the literature. This table is intended for use as a guide until the athlete has developed the neuromuscular attributes that will make testing with heavier loads (e.g., 1RM-5RM) safe and effective. As noted earlier, one should be mindful that the accuracy of the prediction reduces the repetitions' increase (134).

Figure 18.8 A 1RM testing protocol.

1. Instruct the athlete to warm up with a light resistance that easily allows 5 to 10 repetitions.
 2. Provide a 1-minute rest period.
 3. Estimate a warm-up load that will allow the athlete to complete three to five repetitions by adding
 - 10 to 20 pounds (4-9 kg) or 5% to 10% for upper body exercise or
 - 30 to 40 pounds (14-18 kg) or 10% to 20% for lower body exercise.
 4. Provide a 2-minute rest period.
 5. Estimate a conservative, near-maximal load that will allow the athlete to complete two or three repetitions by adding
 - 10 to 20 pounds (4-9 kg) or 5% to 10% for upper body exercise or
 - 30 to 40 pounds (14-18 kg) or 10% to 20% for lower body exercise.
 6. Provide a 2- to 4-minute rest period.
 7. Make a load increase:
 - 10 to 20 pounds (4-9 kg) or 5% to 10% for upper body exercise or
 - 30 to 40 pounds (14-18 kg) or 10% to 20% for lower body exercise
 8. Instruct the athlete to attempt a 1RM.
 9. If the athlete was successful, provide a 2- to 4-minute rest period and go back to step 7. If the athlete failed, provide a 2- to 4-minute rest period; then decrease the load by subtracting
 - 5 to 10 pounds (2-4 kg) or 2.5% to 5% for upper body exercise or
 - 15 to 20 pounds (7-9 kg) or 5% to 10% for lower body exercise.
- AND then go back to step 8.
- Continue increasing or decreasing the load until the athlete can complete one repetition with proper exercise technique. Ideally, the athlete's 1RM will be measured within three to five testing sets.

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Using Prediction Equations Equations are also available to predict the 1RM from xRM loads (103, 104). Researchers who have reviewed such equations report that as the loads used in xRM testing become heavier (i.e., bringing the loads closer to the actual 1RM), the accuracy of the 1RM estimation increases (45). Furthermore, the results obtained from lower xRM testing (and subsequent predictions of the 1RM) are generally more accurate when an athlete has been

consistently training with low xRM resistances (i.e., heavy loads) for a few months before testing (14).

xRM Testing Based on Goal Repetitions

A third option for determining training loads requires the strength and conditioning professional to first decide on the number of repetitions (i.e., the **goal repetitions**) the athlete will perform in the actual program for the exercise being tested. For example, if the strength and conditioning professional decides that the athlete should perform 6 repetitions for the bench press exercise in the training program, the xRM testing protocol should have the athlete perform the exercise with a load that will result in 6 repetitions (6RM). Core and assistance exercises can be selected for xRM testing, but as previously mentioned, high-repetition testing sets can create significant fatigue and may compromise the accuracy of the tested xRM. This effect seems to be more problematic for exercises that involve multiple joints and large muscle areas due to their high metabolic demand (138, 185). Further, xRM testing (and subsequent load assignments) for assistance exercises should be at or above an 8RM to minimize the isolative stress on the involved joint and connective tissue (2, 46). In other words, even if an athlete is following a muscular strength training program that involves 2RM loads for the core exercises, the heaviest load the assistance exercises should be assigned is an 8RM.

TABLE 18.9 Estimating 1RM and Training Loads

Maximum repetitions (RM)	1	2	3	4	5	6	7	8	9	10	12	15
%1RM	100	95	93	90	87	85	83	80	77	75	67	65
Load (pounds or kilograms)	10	10	9	9	9	9	8	8	8	8	7	7
	20	19	19	18	17	17	17	16	15	15	13	13
	30	29	28	27	26	26	25	24	23	23	20	20
	40	38	37	36	35	34	33	32	31	30	27	26
	50	48	47	45	44	43	42	40	39	38	34	33
	60	57	56	54	52	51	50	48	46	45	40	39
	70	67	65	63	61	60	58	56	54	53	47	46
	80	76	74	72	70	68	66	64	62	60	54	52
	90	86	84	81	78	77	75	72	69	68	60	59
	100	95	93	90	87	85	83	80	77	75	67	65
	110	105	102	99	96	94	91	88	85	83	74	72

	120	114	112	108	104	102	100	96	92	90	80	78
	130	124	121	117	113	111	108	104	100	98	87	85
	140	133	130	126	122	119	116	112	108	105	94	91
	150	143	140	135	131	128	125	120	116	113	101	98
	160	152	149	144	139	136	133	128	123	120	107	104
	170	162	158	153	148	145	141	136	131	128	114	111
	180	171	167	162	157	153	149	144	139	135	121	117
	190	181	177	171	165	162	158	152	146	143	127	124
	200	190	186	180	174	170	166	160	154	150	134	130
	210	200	195	189	183	179	174	168	162	158	141	137
	220	209	205	198	191	187	183	176	169	165	147	143
	230	219	214	207	200	196	191	184	177	173	154	150
	240	228	223	216	209	204	199	192	185	180	161	156
	250	238	233	225	218	213	208	200	193	188	168	163
	260	247	242	234	226	221	206	208	200	195	174	169
	270	257	251	243	235	230	224	216	208	203	181	176
	280	266	260	252	244	238	232	224	216	210	188	182
	290	276	270	261	252	247	241	232	223	218	194	189
	300	285	279	270	261	255	249	240	231	225	201	195
	310	295	288	279	270	264	257	248	239	233	208	202
	320	304	298	288	278	272	266	256	246	240	214	208
	330	314	307	297	287	281	274	264	254	248	221	215
	340	323	316	306	296	289	282	272	262	255	228	221
	350	333	326	315	305	298	291	280	270	263	235	228
	360	342	335	324	313	306	299	288	277	270	241	234

Maximum repetitions (RM)	1	2	3	4	5	6	7	8	9	10	12	15
%1RM	370	352	344	333	322	315	307	296	285	278	248	241
Load (pounds or kilograms)	380	361	353	342	331	323	315	304	293	285	255	247
	390	371	363	351	339	332	324	312	300	293	261	254
	400	380	372	360	348	340	332	320	308	300	268	260

Data from Hoeger et al. (1987); Hoeger et al. (1990); Mayhew et al. (1992).

	410	390	381	369	357	349	340	328	316	308	274	267
	420	399	391	378	365	357	349	336	323	315	281	273
	430	409	400	387	374	366	357	344	331	323	288	280
	440	418	409	396	383	374	365	352	339	330	295	286
	450	428	419	405	392	383	374	360	347	338	302	293
	460	437	428	414	400	391	382	368	354	345	308	299
	470	447	437	423	409	400	390	376	362	353	315	306
	480	456	446	432	418	408	398	384	370	360	322	312
	490	466	456	441	426	417	407	392	377	368	328	319
	500	475	465	450	435	425	415	400	385	375	335	325
	510	485	474	459	444	434	423	408	393	383	342	332
	520	494	484	468	452	442	432	416	400	390	348	338
	530	504	493	477	461	451	440	424	408	398	355	345
	540	513	502	486	470	459	448	432	416	405	362	351
	550	523	512	495	479	468	457	440	424	413	369	358
	560	532	521	504	487	476	465	448	431	420	375	364
	570	542	530	513	496	485	473	456	439	428	382	371
	580	551	539	522	505	493	481	464	447	435	389	377
	590	561	549	531	513	502	490	472	454	443	395	384
	600	570	558	540	522	510	498	480	462	450	402	390

Data from Hoeger et al. (1987); Hoeger et al. (1990); Mayhew et al. (1992).

Rating of Perceived Exertion and Repetitions in Reserve

Beyond percentages of 1RM, other methods of load prescription are available to strength and conditioning professionals. The **rating of perceived exertion (RPE)** provides a subjective measurement of work based on the perception of an individual. While the original RPE scale (i.e., 6-20) was based on aerobic activity (15), a modified CR10 scale (i.e., 0-10) has been examined by researchers when monitoring resistance training exercise intensity within an individual set (92, 135) and during entire training sessions (i.e., session RPE) (37, 107, 181). Similarly, **repetitions in reserve (RIR)** provides a subjective measure of the perceived number of repetitions that theoretically could be performed following an exercise set (199). Hackett and colleagues (64, 66)

displayed very large relationships between estimated RIR and actual RIR and showed that RIR estimations become more accurate when exercise sets are performed closer to failure. These findings are supported by additional researchers (100, 198); thus, strength and conditioning professionals must consider that the relative intensity of each exercise set may need to be higher for this method to be useful for load prescription purposes.

When implementing RPE and RIR, it is recommended to prescribe intensities in the form of ranges (e.g., RPE of 7-8 or RIR of 2-3) (176). Although relatively few studies exist, some have shown that load prescription based on RPE for 8 weeks (73) and RIR for 12 weeks (60) may lead to greater strength improvements compared to using percentages of 1RM. However, further studies with additional athlete populations are needed before concrete conclusions can be drawn. RPE and RIR may be best used if combined with an objective method (e.g., percentage of 1RM) so that the desired loads for a given exercise can be determined before using RPE/RIR to adjust the load for autoregulation purposes (176). Professionals should note that the ability to gauge physical exertion may vary based on athletic experience (4). For example, inexperienced athletes may not understand what a near-maximal or maximal resistance training effort feels like. Thus, strength and conditioning professionals should consider limiting load autonomy until the athletes can display proficiency with the RPE and RIR methods.

Set-Repetition Best

Originally proposed by Stone and O'Bryant (162), **set-repetition best (SRB)** is a load prescription method that uses relative intensities (i.e., percent ranges) where the maximum load is estimated based on performances of a given set and repetition scheme. In other words, SRB determines loads based on percentages of what was performed in training. Specifically, 5% ranges are prescribed to allow strength and conditioning professionals to observe the performance of their athletes and incorporate athlete feedback as a means of autoregulation. Beyond the ability to adjust weekly training prescription, SRB allows specific training loads to be estimated when prescribing different set and repetition schemes (e.g., loads for 3 sets of 5 repetitions should be approximately 15% less than 3 sets of 2 repetitions) (162, 176). Compared to prescribing xRM percentages based on single sets, SRB is based on multiple sets so that an athlete's training loads are based on relative percentages of the prescribed set-repetition scheme. Using SRB, any set-repetition scheme can be prescribed and progressed using relative 5% ranges. For example, 3 sets of 5 repetitions or 3 sets of 10 repetitions can be prescribed with 85% to 90% of the respective estimated maximal load for 3 sets of 5 or 3 sets of 10, respectively, in week 1 of

a training phase and progress to 90% to 95% the following week. Within a single week of training, heavy days are typically prescribed with loads approximately 10% to 15% heavier than the lighter days of training (42). Thus, SRB may account for the fatigue that accumulates across multiple sets during training. This may allow training loads to be adjusted on a weekly basis without the need to perform RM testing on a regular basis. [Table 18.10](#) shows SRB relative intensities as they relate to load effort of both RPE and RIR.

TABLE 18.10 Subjective Loading Efforts and Corresponding Set-Repetition Best Relative Percentages

Load effort	Rating of perceived exertion	Repetitions in reserve	Relative percentage
Maximal	10	0 No additional load	100
	9.5	0 Could increase load	
Very heavy	9	1	95-99
	8.5	1-2	
Heavy	8	2	90-94
	7.5	2-3	
Moderate-heavy	7	3	85-89
	6-7	3-4	
Moderate	5-6	4	80-84
		4-5	
Light-moderate	5	5	75-79
Light	4-5	5-6	70-74
Very light	3-4	Light effort	<70
Very, very light	1-2	Little or no effort	<60

Adapted by permission from S. McKeever and R. Howard, "Resistance Training," in *NSCA's Guide to High School Strength and Conditioning*, edited for the National Strength and Conditioning Association by P. McHenry and M. Nitka (Human Kinetics, 2022), 240 and M.C. Zourdos, A. Klemp, C. Dolan, et al., "Novel Resistance Training-Specific Rating of Perceived Exertion Scale Measuring Repetitions in Reserve," *The Journal of Strength and Conditioning Research* 30, no. 1 (2016): 269. Adapted from DeWeese et al. (2014).

A growing body of research supports the use of SRB as a method of load prescription (17, 18, 79, 164, 171-173). Researchers have shown that SRB may produce more favorable muscle architectural (e.g., type I and II hypertrophy and

myosin heavy chain adaptations) and strength (e.g., isometric peak and rapid force production) adaptations compared to RM training (17, 18). Like other methods of load prescription, strength and conditioning professionals should take a conservative approach before progressing to heavier loads. Beyond the initial weeks of training, previous training blocks may be used as starting points for upcoming blocks. For example, the heaviest loads used during a general strength block (e.g., 3 sets of 6 repetitions) may serve as the starting loads for an absolute strength block (e.g., 3 sets of 3 repetitions). Readers are directed to previous literature for more thorough information on SRB (162, 176).

Assigning Load and Repetitions Based on the Training Goal

During the needs analysis, the strength and conditioning professional is challenged to choose the primary goal of the resistance training program based on the athlete's testing results, the movement and physiological analysis of the sport, and the priorities of the athlete's sport season. Once decided on, the training goal can be applied to determine specific load and repetition assignments via the RM continuum and the chosen loading method. As explained previously, the testing methods determine how the loads and repetitions are assigned for each exercise (i.e., loads are calculated as a percentage of a tested or estimated 1RM, or training loads are specifically determined from xRM testing). The options for testing and assigning training loads and repetitions are summarized in [figure 18.9](#).

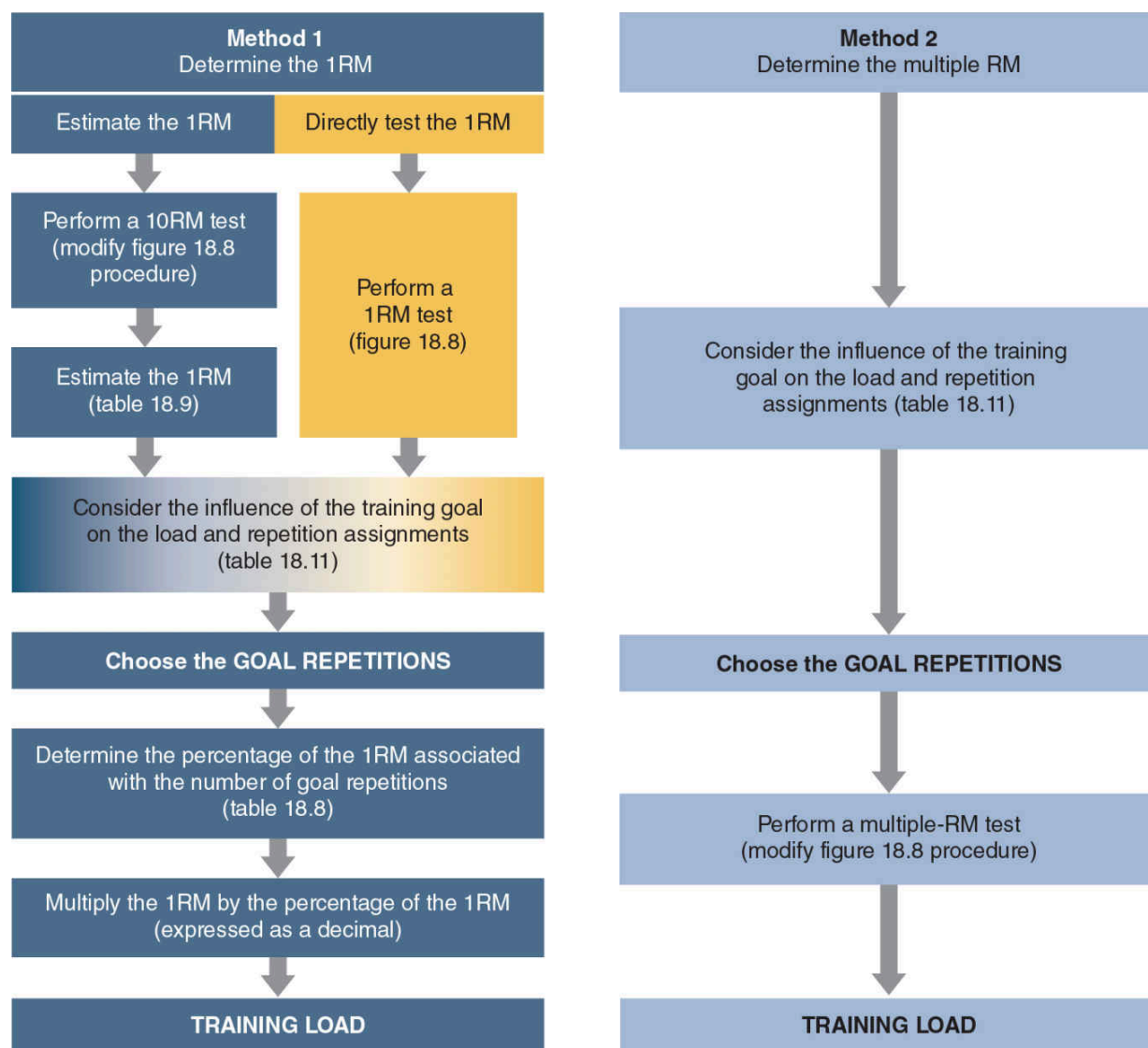


FIGURE 18.9 Summary of testing and assigning training loads and repetitions.

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Repetition Maximum Continuum

[Figure 18.10](#) represents how RM ranges have been historically associated with training goals. In general, relatively heavy loads should be used if the goal is maximal strength, whereas light loads should be used to develop muscular endurance. To state this another way, low xRMs appear to have the greatest effect on maximal strength and high xRMs seem to result in better muscular endurance improvements (65, 106, 141). In contrast to strength and muscular endurance, a spectrum of RM loads may be used to develop muscle hypertrophy. For example, researchers have reported that similar magnitudes of hypertrophy can be produced after eight weeks of training when using either a low-load (25-

35 repetitions performed to failure with approximately 30%-50% 1RM) or high-load (8-12 repetitions performed with approximately 70%-80% 1RM) training protocol that is performed to muscular failure (144). The same research group followed up this study with a meta-analysis that showed no difference in muscle hypertrophy changes following various training protocols performed to muscular failure using lighter (<60% 1RM) and heavier (>60% 1RM) loads (141). As greater muscle activation was shown during multiple heavier versus lighter (i.e., 80% vs. 30% 1RM) sets performed to failure (82), high-load, lower-repetition sets may be considered more efficient at producing hypertrophic adaptations and benefiting muscular strength due to the neural adaptations associated with this characteristic (1). Strength and conditioning professionals should also be cautious in prescribing multiple low-load sets to failure to mitigate potential injury (109).

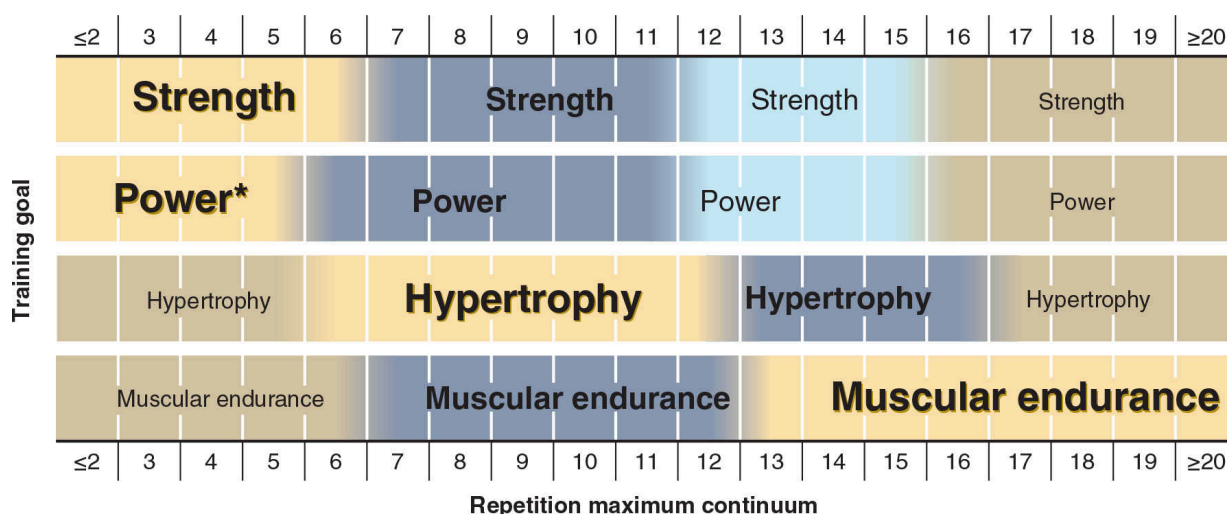


FIGURE 18.10 Classic continuum that shows how RM ranges are associated with various training goals.
 *The repetition ranges shown for power in this figure are not consistent with the %1RM–repetition relationship. Refer to the discussion of assigning percentages for power training for further explanation.

Therefore, while the classic RM continuum effectively illustrates that a certain RM zone *emphasizes* a specific outcome, the training benefits may be blended at any given RM. However, [figure 18.11](#) represents an updated version of the RM continuum that is supported by the emerging literature.

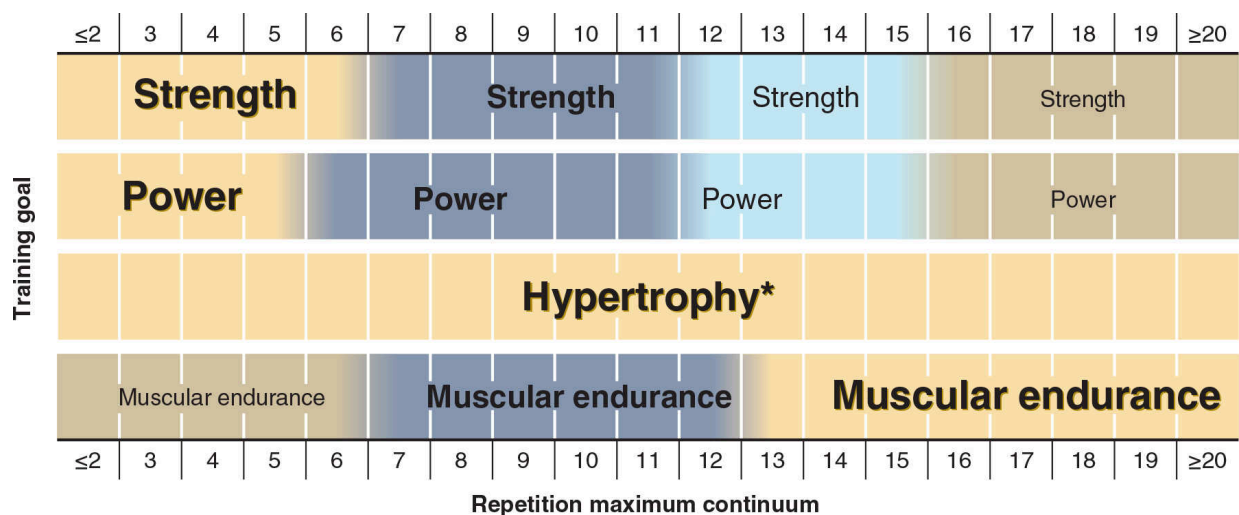


FIGURE 18.11 RM continuum that shows how RM ranges emphasize different training goals while accounting for the expansive range of loads that can be used for hypertrophy.

* The repetition ranges shown for hypertrophy may vary depending on the set and repetition scheme as well as the loads used (141, 187).

Adapted from Stone et al. (1982) and Fleck and Kraemer (2014).

Percentage of the 1RM

Despite the physiological blend of training effects, the specificity principle still dictates the dominant outcome that is attained and enhanced with a particular training load. The relationship between the percentage of the 1RM and the estimated number of repetitions that can be performed at that load ([table 18.8, page 544](#)) allows the strength and conditioning professional to assign a specific resistance to be used for an exercise in a training session. In other words, the training goal is attained when the athlete lifts a load of a certain percentage of the 1RM for a specific number of repetitions ([table 18.11](#)).

TABLE 18.11 Load and Repetition Assignments Based on the Training Goal

Training goal	Load (%1RM)	Goal reps
Strength ^a	≥85	≤6
Multiple-effort ^b event	75-85	3-5

^aThese RM loading assignments for muscular strength training apply only to core exercises; assistance exercises should be limited to loads not heavier than an 8RM (2).

^bBased on weightlifting-derived movements (e.g., clean and snatch). The load and repetition assignments shown for power in this table are *not consistent* with the %1RM–repetition relationship. In nonexplosive movements, loads equaling about 80% of the 1RM apply to the 2- to 5-repetition range. Refer to the discussion of assigning percentages of the 1RM for power training for further explanation.

Data from Berger (1962); Peterson et al. (2004); Peterson et al. (2005); Swinton et al. (2024); Tesch (1992).

Hypertrophy	67-80	8-12
Muscular endurance	≤67	≥12

^aThese RM loading assignments for muscular strength training apply only to core exercises; assistance exercises should be limited to loads not heavier than an 8RM (2).

^bBased on weightlifting-derived movements (e.g., clean and snatch). The load and repetition assignments shown for power in this table are *not consistent* with the %1RM–repetition relationship. In nonexplosive movements, loads equaling about 80% of the 1RM apply to the 2- to 5-repetition range. Refer to the discussion of assigning percentages of the 1RM for power training for further explanation.

Data from Berger (1962); Peterson et al. (2004); Peterson et al. (2005); Swinton et al. (2024); Tesch (1992).

How to Calculate a Training Load If an athlete’s training goal is muscular strength and the tested 1RM in the bench press exercise is 220 pounds (100 kg), to increase strength, the athlete needs to handle loads of at least 85% of the 1RM (after warm-up) that typically allow performance of up to 6 repetitions within a single set ([table 18.11](#)). More specifically, if the strength and conditioning professional assigns 4 repetitions for this exercise, the corresponding load could be approximately 90% of the 1RM ([table 18.8](#)), or approximately 200 pounds (91 kg). Note that the strength and conditioning professional should make adjustments to the assigned loads based on observation of the ease or difficulty an athlete experiences in lifting the load for the required repetitions. Moreover, it is important to avoid training to failure on a regular basis because it may not be necessary to maximize muscular hypertrophy, strength, or power adaptations (36, 63, 137, 191). By minimizing training to failure, strength and conditioning professionals may reduce the overall training strain on athletes while still providing an effective training stimulus and maximizing performance gains (17, 18).

Assigning Percentages for Power Training The force–velocity curve illustrates that the greater the amount of concentric muscular force generated, the slower the muscle shortening and corresponding movement velocity (and vice versa). Maximal power, in contrast, may be produced at intermediate velocities with the lifting of light to moderate loads in nonballistic, traditional resistance training exercises such as the squat or bench press (27, 105, 154, 157). Performing a 1RM involves slower movement velocities; near-maximal forces are generated but with reduced power output (56, 197). Seldom is an athlete required to demonstrate a singular, maximal, slow-speed muscular strength effort in a sport (except in powerlifting, for example). Most sport movements are faster and involve higher power outputs than those produced during a 1RM test (89, 118). This does not mean that an athlete’s power capabilities are unaffected by maximal muscular strength training, however.

Because speed- or power-related sport movements often begin from zero or near-zero velocities, slow-velocity strength gains have direct application to rapid force development and power production (1, 22). For these reasons, the load and repetition assignments for power training overlap the guidelines for strength training ([table 18.11](#)).

Non-weightlifting, multijoint power exercises (e.g., jump squat, bench press throw, overhead press throw) and single-joint muscle action data reveal that peak power is generally reached with the lifting of very light loads—from body weight (0%) to 30% of the 1RM (27, 105, 154, 157, 174, 175). Performing some of these exercises (e.g., bench press throw and overhead press throw) in a Smith machine, for example, can help to address the safety issues. On the other end of the load continuum, data from multiple national- and world-class weightlifting and powerlifting championships clearly indicate that power output increases as the weight lifted decreases from 100% of the 1RM (i.e., the 1RM) to 90% of the 1RM (56). In fact, for the back squat and deadlift exercises, power output for a load at 90% of the 1RM may be *twice* as high as with the 1RM load due to a large decrease in the time required to complete the exercise with the lighter load (57). Even for the already “fast” power exercises (weightlifting-based movements), there is still a 5% to 10% increase in power output as the load decreases from the 1RM to 90% of the 1RM (57). Considering these issues, the most effective and practical application is to assign loads that are about 75% to 90% of the 1RM for resistance training exercises that can be heavily loaded such as the snatch and clean and other weightlifting-derived movements (27, 56, 105). However, the loading that maximizes power output in weightlifting movements and their derivatives is exercise specific. For example, the highest power outputs of catching variations (e.g., hang power clean and power clean) were shown between 60% and 80% 1RM (20, 83, 85, 168), while pulling derivatives may maximize power output at either lighter (i.e., <50% 1RM of a catching variation) (86, 167, 169, 179, 180) or at much heavier loads (i.e., 100%-120% 1RM) (23, 25, 111, 112, 115) depending on the velocity (e.g., jump shrug and hang high pull) or force potential (e.g., countermovement shrug, mid-thigh pull, hang pull) of the exercises, respectively. Strength and conditioning professionals should also not overlook the use of overhead pressing derivatives (e.g., push press and jerk) for developing power output (156). Researchers have shown that maximal power output during the jerk and back jerk occurred between 80% and 90% 1RM of the respective exercises in competitive weightlifters (51). However, strength and conditioning professionals should note that there were no differences in power output or other kinetic variables between the push press, push jerk, and split jerk using 80% of the 1RM push press in resistance-trained men and women (155).

To promote program specificity, particular load and repetition assignments are indicated for athletes training for single-effort power events (e.g., shot put, high jump, weightlifting) and for multiple-effort power events (e.g., basketball and volleyball). For example, single-effort-event athletes may be assigned sets of 1 or 2 repetitions using loads that equal 80% to 90% of the 1RM, especially on heavy training days. For sports with multiple maximum-power efforts (e.g., the frequent maximum vertical jumping motions of a volleyball blocker), 3 to 5 repetitions per set with loads at 75% to 85% of the 1RM may be most appropriate. However, strength and conditioning professionals should consider the training season that the athlete is currently in to ensure that fatigue is being appropriately managed.

On the basis of the %1RM–repetition relationships shown in [table 18.8](#) ([page 544](#)), the strength and conditioning professional may question the load assignments for power training in [table 18.11](#). The %1RM loads may appear to be too low compared to the goal number of repetitions. For example, according to [table 18.8](#), 3 to 5 repetitions are typically associated with loads 93% to 87% of the 1RM, not 75% to 85% of the 1RM or less as [table 18.11](#) indicates. Power exercises cannot be maximally loaded at any repetition scheme, because the quality of the movement technique will decline before momentary muscle fatigue defines a true xRM set (71). Therefore, lighter loads allow the athlete to complete repetitions with maximum speed to promote maximum power development. For example, power exercises are usually limited to 5 repetitions per set, but with loads up to and equal to a 10RM (i.e., approximately 75% of the 1RM). This load adjustment to promote peak power output also applies to the RM continuum ([figure 18.10](#), [page 551](#)). Power training can be emphasized across the range of 5 repetitions or fewer, but the strength and conditioning professional should realize that these loads are not true RMs. However, exercises with a smaller displacement (e.g., countermovement shrug and hang pull) may allow the performance of at least 6 repetitions with supramaximal loads (>100% 1RM power clean) without decreases in force production characteristics or changes in technique across repetitions (113, 114).

Variation of the Training Load

Training for muscular strength and power places a high physiological stress on an athlete's body. Intermediate and advanced resistance-trained athletes are accustomed to lifting heavy loads and possess the experience and motivation to exert to near failure during every set, but this should not always be the goal. Despite the high training status, this degree of training demand typically cannot be tolerated very long without contributing to an overtrained state (176). For example, an athlete may resistance train three days a week with muscular

strength as the goal (e.g., Mondays, Wednesdays, and Fridays). It would be difficult for the athlete to perform the same high-load, low-volume regimen—especially in the power and other core exercises—with only one or two days of rest between sessions. As noted previously, training to failure may not be necessary to promote the desired training adaptations (36, 63, 137, 191).

One strategy to counterbalance the monotony associated with lifting heavy loads is to alter the loads (%1RMs) for the power and other core exercises so that only one training day each week (e.g., Monday) is a heavy day. These heavy-day loads are designed to be high but usually with 1 to 2 RIR to avoid training to failure. The loads for the other training days are reduced (intentionally) to avoid an excessive accumulation of fatigue after the heavy day while still maintaining sufficient training frequency and volume. In the example of the three-days-a-week program, Wednesday can still serve as another heavy training day (depending on the organization of workouts) ([table 18.5, page 538](#)), while Friday may serve as the light day of the week in which 80% of the loads lifted in the power and other core exercises on the heavy day (Monday) are performed with the same number of goal repetitions. Even if the athlete is able to perform more repetitions than the designated goal number, he or she should not do so. This approach can be used for any training frequency. For instance, a two-days-a-week program could have a heavy day and a light day, or a lower body–upper body split routine could consist of two heavy days (one lower body day and one upper body day) followed by two lighter days. Varying the training loads also works well with an athlete’s other training, in that heavy lifting days can fall on light sport conditioning days, and light lifting days on heavy sport conditioning days (42). The strength and conditioning professional needs to monitor this schedule so that it does not lead to heavy training *every* day.

Progression of Load

As the athlete adapts to the training stimulus, the strength and conditioning professional needs to have a strategy for advancing exercise loads (intensity or volume depending on the specific goal) so that improvements will continue over time (progression). Monitoring each athlete’s training and charting his or her response to the prescribed workouts enable the strength and conditioning professional to know when and to what extent the loads should be increased.

Timing Load Increases

A conservative method that can be used to increase an athlete’s training loads is called the **2-for-2 rule** (2). If the athlete can perform 2 or more repetitions over his or her assigned repetition goal for a given exercise in the last set in two consecutive workouts, weight should be added to that exercise for the next

training session. For example, a strength and conditioning professional assigns 3 sets of 6 repetitions in the bench press exercise, and the athlete performs all 6 repetitions in all sets. After several workout sessions (the specific number depends on many factors), the athlete can complete 8 repetitions in the third (last) set for two consecutive workouts. In the following training session, the load for that exercise should be increased. While the 2-for-2 rule may provide strength and conditioning professionals with a tool to progress load in novice athletes, this method may eventually promote training to failure because it does not account for exercise technique, the goals of a given training phase, or relative intensity. For example, if an athlete modifies his or her technique just to achieve additional repetitions, an increase in load may not be warranted until exercise technique becomes more stable. Moreover, programming that allows for additional repetitions to be performed (i.e., not to failure) may be by design (176). In this instance, strength and conditioning professionals may plan for athletes to perform additional repetitions beyond the amount prescribed (i.e., RIR) with the goal of providing an overload stimulus while managing fatigue. Thus, it is advised that the professionals understand the limitations of the 2-for-2 rule when adjusting the load during training.

Quantity of Load Increases

The decision as to the magnitude of the load increase can be difficult to make, but [table 18.12](#) provides general recommendations based on the athlete’s condition (stronger or weaker) and body area (upper or lower body). Despite these guidelines, the significant variation in training status, volume–loads, and exercises (type and muscular involvement) greatly influences the appropriate load increases. To contend with this variability, relative load increases of 2.5% to 10% can be used instead of the absolute values shown in [table 18.12](#). However, load changes should be specific to the exercise, loading type (e.g., free weights, elastic bands, dumbbells), and the method of load progression (e.g., percentage of 1RM, RPE, RIR, SRB).

TABLE 18.12 Examples of Load Increases

Description of the athlete ^a	Body area exercise	Estimated load increase ^b	Set-repetition best increases ^c
Weaker, less trained	Upper body	2.5-5 lb (1-2 kg)	5%

^aThe strength and conditioning professional will need to determine which of these two subjective categories applies to a specific athlete.

^bThese load increases are appropriate for training programs using approximately 3 sets of 5 to 10 repetitions. Note that the goal repetitions per set remain constant as the loads are increased.

^cA planned increase per week, but it will be phase specific, could be larger increases.

	Lower body	5-10 lb (2-4 kg)	5%
Stronger, more trained	Upper body	5-10+ lb (2-4+ kg)	5%
	Lower body	10-15+ lb (4-7+ kg)	5%

ªThe strength and conditioning professional will need to determine which of these two subjective categories applies to a specific athlete.

ºThese load increases are appropriate for training programs using approximately 3 sets of 5 to 10 repetitions. Note that the goal repetitions per set remain constant as the loads are increased.

ºA planned increase per week, but it will be phase specific, could be larger increases.

Application of the Training Load and Repetition Guidelines

Refer to the first scenario table on [page 528](#) for a description of the scenario athletes.

Scenario A Female collegiate basketball player In-season	Scenario B Male professional American football lineman Off-season	Scenario C Male high school cross country runner Preseason
PRIMARY IN-SEASON RESISTANCE TRAINING GOAL Strength/power	PRIMARY OFF-SEASON RESISTANCE TRAINING GOAL Hypertrophy	PRIMARY PRESEASON RESISTANCE TRAINING GOAL Muscular endurance
TESTING AND ASSIGNING LOADS AND REPS Influence of the training goals <i>Power exercises:</i> 75%-85% of the 1RM; 3-5 reps ^a <i>Other core exercises:</i> >85% of the 1RM; <6 reps <i>Assistance exercises:</i> limited to loads not heavier than an 8RM	TESTING AND ASSIGNING LOADS AND REPS Influence of the training goals 67%-80% of the 1RM; 8-12 reps	TESTING AND ASSIGNING LOADS AND REPS Influence of the training goals <67% of the 1RM; >12 reps
Number of goal reps <i>Power exercises:</i> 3 <i>Core exercises:</i> 4 <i>Assistance exercises:</i> 8	Number of goal reps <i>Core exercises:</i> 10 <i>Assistance exercises:</i> 10	Number of goal reps <i>Core exercises:</i> 12 <i>Assistance exercises:</i> 15
Testing methods 3RM testing for power exercises^b Push jerk	Testing methods 1RM testing for core exercises Deadlift	Testing methods 6RM testing for core exercises

Hang power clean 1RM testing for other core exercises^c Front squat Incline bench press Seated row 8RM testing for assistance exercises^d Dumbbell alternating curl Triceps pushdown	Back squat ⁱ Bench press ⁱ Shoulder press ⁱ 10RM testing for new assistance exercises^j Step-up Seated calf (heel) raise Bent-over row Shoulder shrug	Hexagonal barbell deadlift Lunge Vertical chest press 8RM testing for new assistance exercises One-arm dumbbell row Lateral shoulder raise
Testing results 3RM hang power clean 115 lb (52 kg) <i>Estimated 1RM^e</i> 124 lb (56 kg) 3RM push jerk 110 lb (50 kg) <i>Estimated 1RM^e</i> 118 lb (54 kg) 1RM front squat 185 lb (84 kg) 1RM incline bench press 100 lb (45 kg) 1RM seated row 120 lb (54 kg) 8RM dumbbell alternating curl 25 lb (11 kg) 8RM triceps 50 lb (23 kg) pushdown	Testing results 1RM deadlift 650 lb (295 kg) 1RM back squat 675 lb (306 kg) 1RM bench press 425 lb (193 kg) 1RM shoulder press 255 lb (116 kg) 10RM step-up 205 lb (93 kg) 10RM seated calf (heel) raise 155 lb (70 kg) 10RM bent-over row 215 lb (98 kg) 10RM shoulder shrug 405 lb (184 kg)	Testing results 6RM hexagonal barbell deadlift 135 lb (61 kg) <i>Estimated 1RM^e</i> 160 lb (73 kg) 6RM lunge 100 lb (45 kg) <i>Estimated 1RM^e</i> 120 lb (54 kg) 6RM vertical chest press 85 lb (39 kg) <i>Estimated 1RM^e</i> 100 lb (45 kg) 8RM one-arm dumbbell row 35 lb (16 kg) 8RM lateral shoulder raise 15 lb (7 kg)
Training loads <i>For power exercises:</i> Assign 75% of the estimated 1RM Push jerk 90 lb (41 kg) Hang power clean 95 lb (43 kg) (All loads are rounded off to the nearest 5 lb.) <i>For other core exercises:</i> Assign 85% of the tested 1RM Front squat 155 lb (70 kg) Incline bench press 85 lb (39 kg) Seated row 120 lb (54 kg)	Training loads <i>For core exercises:</i> Assign 70% of the tested 1RM Deadlift 490 lb (222 kg) Back squat 505 lb (229 kg) Bench press 320 lb (145 kg) Shoulder press 190 lb (86 kg) (All loads are rounded off to the nearest 5 lb.) <i>For assistance exercises:</i> Assign loads less than the loads from the 10RM testing ^g or	Training loads <i>For core exercises:</i> Assign <67% of the estimated 1RM <i>For assistance exercises:</i> Assign loads less than the loads from 8RM testing ^g

(All loads are rounded off to the nearest 5 lb.) <i>For assistance exercises:</i> Assign loads less than the loads from 8RM testing^g	Equal to the loads used in the postseason Leg (knee) curl 190 lb (86 kg) Barbell biceps curl 115 lb (52 kg) Lying triceps 125 lb (57 kg) extension	
Weekly loading regimen (power/core exercises)^f <i>Mondays ("heavy" push day)</i> Assign the full load assignments (calculated under "Training loads") <i>Wednesdays ("heavy" pull day)</i> Assign the full load assignments (calculated under "Training loads") <i>Fridays ("light" mix day)</i> Assign only 80% of Monday's and Wednesday's "heavy day" loads if exercises are repeated^h	Weekly loading regimen includes a heavy and light emphasis on the lower and upper body days <i>Mondays ("Heavy" lower body day)</i> <i>Tuesdays ("Heavy" upper body day)</i> <i>Thursdays ("Light" lower body day)</i> <i>Fridays ("Light" upper body day)</i> Assign 80%-85% of the heavy day loads during the light day training sessions^h	
COMMENTS ^aThe load and rep assignments shown for power exercises are based on basketball, a multiple-effort event, and are not consistent with the %1RM–repetition relationship. ^bTo test for power, an xRM (3RM) protocol is used. From the result, the 1RM is estimated and load assignments are made by calculating a percent of the estimated 1RM. ^cThe athlete performed these exercises in the preseason, but to raise the accuracy of the load assignments for the in-season, the athlete will be tested to determine the current 1RM. ^dEven though some of these exercises were part of the preseason program, they all require xRM testing because the in-season goal reps for these exercises is 8, rather than 10 from the previous season. ^eEstimate the 1RM using table 18.9. ^fThe loads for the assistance exercises remain constant throughout the week; only the loads for the power and other core exercises change. ^gPrescribing loads that are submaximal to the xRM testing will help athletes avoid consistent training to failure.	COMMENTS ⁱThe athlete performed these exercises in the postseason, but to raise the accuracy of the load assignments for the off-season, the athlete will be tested to determine the current 1RM. ^jThe exercises shown here were selected as new exercises for the off-season program and therefore require 10RM testing. The other assistance exercises were carried over from the postseason and do not require testing because the load and rep assignments will be identical.	

^hPrescribing heavy and light loads with the same exercises during the week may help manage fatigue and prevent consistently training to failure. Even if the athlete is able to, do not allow more reps than the designated goal number (power, 3; other core, 4).

Step 6: Volume

Volume-load relates to the total amount of weight lifted in a training session, and a **set** is a group of repetitions sequentially performed before the athlete stops to rest. Repetition-volume is the total number of repetitions performed during a workout session, and volume-load is the total number of sets multiplied by the number of repetitions per set, then multiplied by the weight lifted per repetition. For example, the volume-load for 2 sets of 10 repetitions with 50 pounds (23 kg) would be expressed as $2 \times 10 \times 50$ pounds or 1,000 pounds (454 kg). (If different sets are performed with different amounts of weight, the volumes per set are calculated and then added to obtain the total training session volume.)

In running exercise, the common (rep) volume measure is distance. If an intensity value is known or measured (e.g., running pace, which relates to percent $\dot{V}O_2\text{max}$), total metabolic energy cost (which is proportional to mechanical work done) can be calculated. This value is comparable to volume-load in resistance exercise. The same concepts are applicable to the number of foot or hand contacts (volume) in plyometric exercise (note that the intensity is not considered when counting contacts but should be considered carefully by the coach), the number of strokes (volume) in swimming or rowing, or the number of throws or jumps (volume) for various sport activities.

Single Versus Multiple Sets

Single-set training may be appropriate for untrained individuals (50) or during the first several months of training (61), but many studies indicate that higher volumes are necessary to promote further gains in strength, especially for intermediate and advanced resistance-trained athletes (126, 177, 182). For example, performing 3 sets of 10 repetitions *without going to failure* enhances strength better than one set to failure in 8 to 12 repetitions (90), although the higher training volume with the use of 3 sets is a contributing factor (50, 162). Therefore, an athlete who performs multiple sets from the initiation of their resistance training program will increase muscular strength faster than with single-set training (90). The strength and conditioning professional cannot expect, however, that an athlete will be able to successfully complete multiple sets with full RM loads at fixed repetition schemes for every exercise in each training session. Fatigue will affect the number of repetitions that can be performed in later sets (116, 150), which is why a RIR approach may be advantageous. It should also be noted that training to failure is not necessary for optimal adaptations (36, 63, 137, 191). Refalo and colleagues (132) also demonstrated that the closer the proximity to failure, the greater the magnitude

of fatigue over the subsequent 24 hours, which should be carefully considered in relation to the additional training planned for that period. As such, this also highlights a limitation of the RIR approach, in that it has been clearly demonstrated that the further away from failure, the less accurate the RIR approach is (6, 70).

Training Status

The training status of athletes affects the volume they will be able to safely tolerate. It is appropriate for an athlete to perform only 1 or 2 sets as a beginner and to add sets as he or she becomes better trained. As the athlete adapts to a consistent and well-designed program, more sets can be added gradually to match the guidelines associated with the given training goal.

Primary Resistance Training Goal

Training volume is directly based on the athlete's resistance training goal. [Table 18.13](#) provides a summary of the guidelines for the number of repetitions and sets commonly associated with strength, power, hypertrophy, and muscular endurance training programs. When programming weightlifting exercises and their derivatives, loads and the associated repetition schemes may vary from other forms of power training, because weightlifting pulling derivatives (i.e., those that exclude the catch phase) permit loads greater than the 1RM associated with the catching variation (21). In addition, reduced displacement during some of these exercises (i.e., countermovement shrug, hang pull, power clean) also permits the performance of 6 repetitions without decreases in velocity of power across repetitions (113, 114) ([table 18.14](#)).

TABLE 18.13 Volume Assignments Based on the Training Goal

Training goal	Goal repetitions			Sets ^a		
	NOVICE	INTERMEDIATE	ADVANCED	NOVICE	INTERMEDIATE	ADVANCED
Strength	≤6	≤6	≤6	1-3	≥3	≥3
Power: ^b	n/a	1-3	1-3	n/a	1-3	3-6

^aThese assignments do not include warm-up sets and typically apply to core exercises only.

^bBased on weightlifting movements (clean, snatch, jerk) and their derivatives. The load and repetition assignments shown for power in this table are *not consistent* with the %1RM–repetition relationship. In nonexplosive movements, loads equaling about 80% of the 1RM apply to the 2- to 5-repetition range. Refer to the discussion of assigning percentages of the 1RM for power training for further explanation.

^cA higher number of repetitions may be performed without changes in velocity or power if the range of motion and complexity is reduced (113, 114), or if cluster sets are used.

^dA low number of repetitions for hypertrophy should be mixed with a higher number of sets (e.g., 10 sets of 3 repetitions).

Data from Fleck and Kraemer (2014); Stone and O'Bryant (1987); Tesch (1992).

Single-effort event	n/a	3-6 ^c	3-6 ^c	n/a	1-3 ^c	3-6 ^c
Multiple-effort event						
Hypertrophy	8-12	6-12	3-12	1-3	≥3	≥3 ^d
Muscular endurance	12-15	12-15	12-25	1-3	≥3	≥3

^aThese assignments do not include warm-up sets and typically apply to core exercises only.

^bBased on weightlifting movements (clean, snatch, jerk) and their derivatives. The load and repetition assignments shown for power in this table are *not consistent* with the %1RM–repetition relationship. In nonexplosive movements, loads equaling about 80% of the 1RM apply to the 2- to 5-repetition range. Refer to the discussion of assigning percentages of the 1RM for power training for further explanation.

^cA higher number of repetitions may be performed without changes in velocity or power if the range of motion and complexity is reduced (113, 114), or if cluster sets are used.

^dA low number of repetitions for hypertrophy should be mixed with a higher number of sets (e.g., 10 sets of 3 repetitions).

Data from Fleck and Kraemer (2014); Stone and O'Bryant (1987); Tesch (1992).

Strength and Power

In classic research, DeLorme (39) and DeLorme and Watkins (40) recommended sets of 10 repetitions as ideal to increase muscular strength, although the regimen was originally developed for injury rehabilitation. Later, Berger (9, 10) determined that 3 sets of 6 repetitions created maximal strength gains, at least in the bench press and back squat exercises. Although Berger's work seemed to be conclusive, his subsequent research (11) showed no significant difference among 6 sets of a 2RM load, 3 sets of a 6RM load, and 3 sets of a 10RM load, despite the differences in volume. Since then, the results of many other studies have also been unable to support an exact set and repetition scheme to promote maximal increases in strength; however, these inconclusive findings may be the result of a mixture of relatively untrained and trained subjects, as well as varying programming tactics.

When training an athlete for strength, assigning volume begins with an examination of the optimal number of repetitions for maximal strength gains. As discussed earlier (and shown in [figure 18.10](#) and [table 18.11](#)), this appears to be sets of 6 or fewer repetitions (at the corresponding RM load [≥85% 1RM]) for core exercises (125, 182). Peterson and colleagues (125, 126) also highlighted a dose–response relationship based on the number of sets performed per muscle group: 8 sets per muscle group elicited the greatest increases in strength, with loads of 85% of the 1RM. More recently, similar conclusions were made by Swinton and colleagues (182) based on the results of their meta-analysis of the dose–response to resistance training. Specific set guidelines based on exercise

type suggest that only 1 to 3 sets may be appropriate or necessary for assistance exercises (2).

Volume assignments for power training are typically lower than those for strength training to maximize the quality of exercise. This reduction in volume results from fewer goal repetitions and lighter loads ([figure 18.10](#) and [table 18.11](#)) rather than the recommended number of sets (30, 68, 105). The common guideline is 3 to 5 sets (after warm-up) for power exercises included in a trained athlete's program (161).

TABLE 18.14 Weightlifting Exercises and Derivatives, Including Recommended Intensities and Repetition Ranges

Including Recommended Intensities and Repetition Ranges					
Catching derivatives ^a	Load (reps)	Pulling derivatives ^b	Load (reps)	Overhead pressing derivatives	Load (reps)
Mid-thigh clean; snatch	60%-80% 1RM (2-5)	Mid-thigh pull; countermovement shrug	100%-140% 1RM (4-6 ^{c,d})	Push press	70%-90% 1RM (2-5)
Clean; snatch from the knee	70%-90% 1RM (2-5)	Pull from the knee; hang pull		Push jerk	
Hang clean; snatch ^c		Pull from the floor	80%-120% 1RM (2-6 ^d)	Split jerk	
Clean; snatch		Hang high pull	45%-60% 1RM (4-6)	Behind the neck push press/jerk ^b	
		Jump shrug	30%-45% 1RM (4-6)	Behind the neck split jerk ^b	

^aAll clean and snatch variations can be performed with a partial squat (power) or full squat catch.

^bThe derivatives may be performed with either a clean or snatch grip.

^cSix reps can be performed with no decrease in velocity or power between reps (113, 114).

^dHigher reps may be performed provided that the technique of the exercise is maintained (91, 141).

Note: Variations from mid-thigh and the knee can start with the barbell resting on blocks, or with the athlete holding the barbell and lowering to the start position and briefly pausing. Currently, there is minimal research comparing the kinetics or kinematics of these variations.

Adapted from Comfort et al. (2023). Data from Kirksey and Stone (1998); Scala et al. (1987).

Hypertrophy

It is generally accepted that higher training volumes are associated with increases in muscular size (96, 139, 143). This is the result of both a moderate to higher number of repetitions per set (8-12; see [figure 18.10](#) and [table 18.11](#)) and the commonly recommended 3 to 6 sets per exercise (91, 123). However, there does not appear to be a meaningful difference in training-induced hypertrophy of a specific muscle group between higher and lower frequencies

when volume is matched (140). Furthermore, it has been reported that there are no differences in hypertrophic adaptations following low (<60% 1RM) or high (>60% 1RM) load training programs (141). Thus, it is important that strength and conditioning professionals understand that heavy loads (i.e., >80% 1RM) can be used to train muscle hypertrophy rather than only maximal strength characteristics. In fact, the use of heavier loads may target specific fibers and higher threshold motor units (i.e., type II) to a greater extent than lighter loads (82, 106, 158), which in turn may allow for greater force, rapid force, and power output adaptations in future training blocks (187). Thus, it has been suggested that using heavier loads to target hypertrophic adaptations may be more beneficial to strength and power athletes (53, 97). However, it is also important to adjust the number of sets and repetitions compared to traditional hypertrophy prescription; for example, strength and conditioning professionals could prescribe 10 sets of 3 repetitions when using heavier loads compared to the more traditional 3 sets of 10 repetitions typically prescribed when targeting hypertrophy. Moreover, caution should be taken when prescribing training in this manner because the volume–loads can be quite high and the time it may take to complete a given exercise with proper rest periods between sets may lead to lengthy training sessions. Therefore, it is suggested that users do not implement this set and repetitions scheme with every exercise.

Muscular Endurance

Resistance training programs that emphasize muscular endurance involve performing many repetitions—12 or more—per set (160, 161). Despite this relatively high repetition assignment, the overall volume–load is not necessarily overly inflated since the loads lifted are lighter and fewer sets are performed, commonly 2 or 3 per exercise.

Application of the Volume Guidelines

Refer to the first scenario table on [page 528](#) for a description of the scenario athletes.

Scenario A Female collegiate basketball player In-season	Scenario B Male professional American football lineman Off-season	Scenario C Male high school cross country runner Preseason
Power exercises 3 sets of 3 reps Other core 3 sets of 4 reps exercises Assistance 2 sets of 8 reps exercises (The number of sets does not include warm-ups.)	Core exercises 3 sets of 10 reps Assistance 3 sets of 10 reps exercises (The number of sets does not include warm-ups.)	Core exercises 3 sets of 12 reps Assistance 2 sets of 15 reps exercises (The number of sets does not include warm-ups.)

MONDAY Push jerk 3 × 3 ^a Front squat 3 × 4 Incline bench press 3 × 4 Triceps pushdown 3 × 8 WEDNESDAY Hang power clean 3 × 3 Romanian deadlift 3 × 4 Seated row 3 × 4 Dumbbell 3 × 8 alternating curl FRIDAY Jump shrug 3 × 3 Rear leg—elevated 3 × 4 split-squat Pull-up 3 × 4 Single-leg stiff- 3 × 4 legged deadlift	LOWER BODY (MONDAY AND THURSDAY) Deadlift 3 × 10 Back squat 3 × 10 Step-up 3 × 10 Leg (knee) curl 3 × 10 Seated calf (heel) 3 × 10 raise UPPER BODY (TUESDAY AND FRIDAY) Bench press 3 × 10 Bent-over row 3 × 10 Shoulder press 3 × 10 Barbell biceps curl 3 × 10 Shoulder shrug 3 × 10 Lying triceps 3 × 10 extension Abdominal crunch 3 × 20	WEDNESDAY Hexagonal barbell 3 × 12 deadlift Vertical chest press 3 × 12 One-arm dumbbell 3 × 12 row Abdominal crunch 2 × 20 Machine back 2 × 15 extension SATURDAY Lunge 3 × 12 Lateral shoulder 3 × 12 raise Single-leg stiff- 3 × 12 legged deadlift Toe raise 3 × 12 (dorsiflexion) Abdominal crunch 2 × 20 Machine back 2 × 15 extension
COMMENTS ^a Represented as sets × repetitions here and in scenarios A, B, and C.		

Step 7: Rest Periods

The time dedicated to recovery between sets and exercises is called the **rest period** or **inter-set rest**. The length of the rest period between sets and exercises is highly dependent on the goal of training, the relative load lifted, and the athlete's training status. (If the athlete is not in good physical condition, rest periods initially may need to be longer than typically assigned.)

The amount of rest between sets is strongly related to load; the heavier the loads lifted, the longer the rest periods the athlete will need between sets to safely and successfully complete the prescribed subsequent sets and to promote higher quality training. For example, training for muscular strength with 4RM loads would likely require significantly longer rest periods between sets than training for muscular endurance in which lighter 15RM loads are lifted. Despite the relationship between training goals and the length of rest periods (e.g., long rest periods for muscular strength training programs), not all exercises in a resistance training program should be assigned the same rest periods. It is important that the strength and conditioning professional allocate rest periods based on the relative load lifted and the amount of muscle mass involved in each exercise. An example of this specificity is for an assistance exercise as part of a muscular strength training program. Whereas a core exercise such as the bench press may involve a 4RM load and a 4-minute rest period, an assistance exercise such as the lateral shoulder raise may be performed with a 12RM load and therefore require only a 1-minute rest period (even though 1-min rest periods generally apply to a hypertrophy training program). The recommended rest period lengths for strength, power, hypertrophy, and muscular endurance programs are shown in [table 18.15](#).

TABLE 18.15 Rest Period Length Assignments Based on the Training Goal

Training goal*	Rest period length
Strength	2-5 min
Power (single- and multiple-effort event)	2-5 min

*Because there are occasions when the prescribed percentage of the 1RM for assistance exercises falls outside the range associated with the training goal (e.g., ≥ 8 RM loads are recommended for assistance exercises as part of a muscular strength training program [2]), the strength and conditioning professional should examine the loads used for each exercise when assigning rest periods rather than generally applying the guidelines for a training goal.

Data from Fleck and Kraemer (2014); Gettman and Pollock (1981); Robinson et al. (1995); Scala et al. (1987); Schoenfeld et al. (2016); Stone and O'Bryant (1987); Stone et al. (2022); Tesch et al. (1986).

Hypertrophy	2-3 min for multijoint exercises; 60-90 s for single-joint exercises
Muscular endurance	≤30 s

*Because there are occasions when the prescribed percentage of the 1RM for assistance exercises falls outside the range associated with the training goal (e.g., ≥8RM loads are recommended for assistance exercises as part of a muscular strength training program [2]), the strength and conditioning professional should examine the loads used for each exercise when assigning rest periods rather than generally applying the guidelines for a training goal.

Data from Fleck and Kraemer (2014); Gettman and Pollock (1981); Robinson et al. (1995); Scala et al. (1987); Schoenfeld et al. (2016); Stone and O'Bryant (1987); Stone et al. (2022); Tesch et al. (1986).

Strength and Power

Training may enhance an athlete's ability to exercise with less rest, but athletes who seek to perform maximal or near-maximal repetitions with a heavy load usually need long rest periods, especially for lower body or all-body structural exercises (149, 193). For example, Robinson and colleagues (136) observed that, in the back squat exercise, 3 minutes of inter-set rest resulted in greater strength gains than a 30-second rest period. Common guidelines for rest period length range 2 to 5 minutes (50, 162, 163) because these rest periods have been shown to permit greater volumes of work (194), allow for the use of heavier loads (193), and result in greater maximal strength and power adaptations (38, 136) compared to shorter rest intervals.

Hypertrophy

Athletes who are interested in gaining muscular size often use a short to moderate inter-set rest period (50). In fact, earlier reviews of hypertrophy training programs supported the use of shorter rest periods because it was recommended that the athlete begin the next set before full recovery has been achieved (72, 184). Despite this, the high metabolic demand of exercises involving large muscle groups merits consideration (i.e., extra recovery time) when rest period lengths are being assigned (138, 185). While rest periods for hypertrophy are typically 1.5 minutes or less, longer rest periods (i.e., 3 min) have been shown to produced greater increases in muscle hypertrophy compared to those that were shorter (i.e., 1 min) (145). Longer rest periods may allow athletes to train with a higher overall volume-load that may coincide with the use of heavier training loads (62), which may in turn increase time under tension and the stimulus for hypertrophy. Thus, rest periods for training muscle hypertrophy may exist on a spectrum depending on the loads used and the targeted mechanism of hypertrophy (i.e., tension, metabolic alterations, or damage).

Muscular Endurance

A muscular endurance training program has very short rest periods, often less than 30 seconds. This restriction of the recovery time is purposeful; only a minimal amount of rest is allowed when light loads are being lifted for many repetitions. This type of program is designed to meet the guideline of the specificity principle for muscular endurance (2). Short rest periods are characteristic of circuit training programs (58) in which it is common to alternate exercises and limit rest period lengths to 30 seconds or less.

Conclusion

Well-designed programs are based on the application of sound principles during each step of a process referred to as *program design*. The process begins with a needs analysis to determine the specific demands of the sport and the training status of the athlete. With this knowledge, appropriate exercises are selected, and training frequency is established. The order of exercises in the workout is considered next, followed by load assignments and training volume choices based on desired training outcomes. Deciding on the length of the rest periods is the last step leading to the design of a sport-specific resistance training program. A composite view that includes all the program design variables (steps 1-7) for the three scenarios is shown in the scenario table.

Application of the Rest Period Guidelines

Refer to the first scenario table on [page 528](#) for a description of the scenario athletes.

Scenario A Female collegiate basketball player In-season	Scenario B Male professional American football lineman Off-season	Scenario C Male high school cross country runner Preseason
Power and core 3 min exercises Assistance exercises 60-90 s	Core exercises 2-3 minutes Assistance exercises 60-90 seconds	Core exercises 30 s Assistance exercises 20 s ^d
MONDAY Push jerk 3 min Front squat 3 min Incline bench press 3 min Triceps pushdown 60-90 s ^a WEDNESDAY Hang power clean 3 min Romanian deadlift 3 min Seated row 3 min Dumbbell 60-90 s ^a alternating curl FRIDAY Jump shrug 3 min Rear leg-elevated 3 min split-squat Pull-up 3 min Single-leg stiff- 3 min legged deadlift	LOWER BODY (MONDAY AND THURSDAY) Deadlift 2-3 min Back squat 2-3 min Step-up 2-3 min ^b Leg (knee) curl 60-90 s Seated calf (heel) 60-90 s raise UPPER BODY (TUESDAY AND FRIDAY) Bench press 2-3 min Bent-over row 60-90 s Shoulder press 2-3 min Barbell biceps curl 60-90 s Shoulder shrug 60-90 s Lying triceps 60-90 s extension Abdominal crunch 20 s ^c	WEDNESDAY Hexagonal barbell 30 s deadlift Vertical chest press 30 s One-arm dumbbell 30 s row Abdominal crunch ≤20 s Machine back ≤20 s extension SATURDAY Lunge 30 s Lateral shoulder 30 s raise Single-leg stiff- 30 s legged deadlift Toe raise 30 s (dorsiflexion) Abdominal crunch ≤20 s

		Machine back extension ≤20 s
COMMENTS ^a Despite following a strength/power training program, the athlete is performing sets of 8 reps in this exercise, a volume assignment for hypertrophy training. Therefore, the length of the rest period should be 60-90 s. The rest period for the single-joint exercises is slightly shorter because fewer muscles are involved.	COMMENTS ^b This exercise is classified as an assistance exercise and, like the others, could be assigned a 60-90 s rest period. Despite this, the step-up is a unilateral exercise that requires more time for completion of each set. Therefore, a longer rest period is provided. ^c Although this athlete is training for hypertrophy, he is performing sets of 20 reps in this exercise, a volume assignment for muscular endurance. Therefore, the length of the rest period should be ≤30 s.	COMMENTS ^d Despite the short rest periods used during a muscular endurance phase, shorter rest periods can be used for assistance exercises due to a decreased neuromuscular demand compared to core exercises.

Application of All Program Design Variables: Steps 1-7

Refer to the first scenario table on [page 528](#) for a description of the scenario athletes.

Scenario A Female collegiate basketball player In-season	Scenario B Male professional American football lineman Off-season	Scenario C Male high school cross country runner Preseason
MONDAY (HEAVY PUSH DAY) Push jerk ^a 3 × 3 @ 90 lb (41 kg) Front squat ^a 3 × 4 @ 155 lb (70 kg) Incline bench press ^a 3 × 4 @ 85 lb (39 kg) Triceps pushdown ^b 3 × 8 @ 40 lb (18 kg) WEDNESDAY (HEAVY PULL DAY) Hang power clean ^a 3 × 3 @ 95 lb (43 kg) Romanian deadlift ^a 3 × 4 @ 155 lb (70 kg)	LOWER BODY (MONDAY AND THURSDAY)^d Deadlift ^a 3 × 10 @ 345 lb (156 kg) Back squat ^a 3 × 10 @ 355 lb (161 kg) Step-up ^a 3 × 10 @ 190 lb (86 kg) Leg (knee) curl ^b 3 × 10 @ 175 lb (79 kg) Seated calf (heel) raise ^b 3 × 10 @ 145 lb (66 kg) UPPER BODY (TUESDAY AND FRIDAY)^d Bench press ^a 3 × 10 @ 225 lb (102 kg) Bent-over row ^a 3 × 10 @ 205 lb (93 kg) Shoulder press ^a 3 × 10 @ 135 lb (61 kg) Barbell biceps curl ^b 3 × 10 @ 105 lb (48 kg) Shoulder shrug ^b 3 × 10 @ 375 lb (170 kg) Lying triceps extension ^b 3 × 10 @ 110 lb (50 kg)	WEDNESDAY Hexagonal barbell deadlift ^e 3 × 12 @ 90 lb (41 kg) Vertical chest press ^e 3 × 12 @ 55 lb (25 kg) One-arm dumbbell row ^e 3 × 12 @ 20 lb (9 kg) Abdominal crunch ^f 3 × 20 Machine back extension ^f 3 × 15 @ 45 lb (20 kg) SATURDAY Lunge ^e 3 × 12 @ 65 lb (29 kg) Lateral shoulder raise ^f 3 × 12 @ 10 lb (5 kg)

Seated row ^a 3 × 4 @ 120 lb (54 kg) Dumbbell alternating curl ^b 3 × 8 @ 20 lb (9 kg) FRIDAY (LIGHT MIX DAY) Jump shrug ^a 3 × 3 @ 35 lb (16 kg) ^c Rear leg–elevated split-squat ^a 3 × 4 @ 75 lb (34 kg) Pull-up ^a 3 × 4 @ body weight Single-leg stiff-legged deadlift 3 × 4 @ 35 lb (16 kg) ^a	Abdominal crunch ^e 3 × 20	Single-leg stiff-legged deadlift ^e 3 × 12 10 lb (5 kg) Toe raise (dorsiflexion) ^f 3 × 12 @ 20 lb (9 kg) Abdominal crunch ^f 3 × 20 Machine back extension ^f 3 × 15 @ 40 lb (18 kg)
COMMENTS ^a 3 min rest period ^b 60-90 s rest period ^c Load based on approximately 30% of the hang power clean estimated 1RM	COMMENTS ^d 80%-85% of the loads used on Monday and Tuesday should be prescribed for the training sessions on Thursday and Friday, respectively ^e 30 s rest period	COMMENTS ^f 20 second rest period

KEY TERMS

1-repetition maximum (1RM)
2-for-2 rule
agonist
antagonist
assistance exercise
circuit training
compound set
core exercise
exercise history
exercise order
exercise selection
exercise technique experience
goal repetitions
individualization
injury analysis
intensity
inter-set rest
load
mechanical work
microdosing
movement analysis

multijoint exercise
multiple-RM (xRM)
muscle balance
needs analysis
overload
physiological analysis
power exercise
profile
program design
progression
rating of perceived exertion (RPE)
repetition maximum (RM)
repetition-volume
repetitions
repetitions in reserve (RIR)
rest period
reversibility
SAID
set
set-repetition best (SRB)
single-joint exercise
specificity
split routine
structural exercise
superset
training background
training frequency
training status
variation
velocity
volume
volume-load
volume-load displacement

STUDY QUESTIONS

1. A basketball coach informs you that one of her athletes needs to increase her vertical jump performance. In addition to some ballistic and plyometric training, which of the following resistance training exercises are most likely to enhance the athlete's vertical jump performance?